ITA0522- COMPUTER VISION

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CIA-1

1) Blister-pack Inspection pipeline.

1. Image acquisition: CCD / CMOS camera captures blister pack images.
2. Image Sensing: converts reflected light into electrical signals.
3. Illumination: LED lighting provides uniform brightness
4. Optical components: Lens focuses light onto the camera sensor.
5. Image formation: Forms the tablet image on the sensor.
6. Sampling: converts the image into discrete pixels.
7. Quantization: Assigns intensity values (0-255) to pixels.
8. Image enhancement: Assign numerical intensity values to each pixel, such as 0-255 or an 8-bit.
9. Defect detection: Image processing techniques identify missing tablets, cracks and colour changes.
10. Final Inspection: The system identifies defective blister packs and automatically rejects them.

2) Ceramic tile image enhancement.

Given pixel Intensities: 100, 110, 120, 140, 185.

a) Contrast Stretching

$$S = \frac{r - r_{min}}{r_{max} - r_{min}} \times 255$$

Input	calculation	output
100	(100-100)/85×255	0
110	(110-100)/85×255	30
120	(120-100)/85×255	60
140	(140-100)/85×255	120
185	(185-100)/85×255	255

b) Original mean:

$$= \frac{100+110+120+140+185}{5} = \frac{655}{5}$$

$$= 131$$

Enhanced Mean:-

$$= \frac{0+30+60+120+255}{5} = \frac{465}{5}$$

$$= 93$$

c) Additional operations:-

1. Median filtering

* Removes salt and pepper noise.
* preserve important edges.

2. Unsharp masking

* Enhances fine details
* Makes paint cracks more visible.

3) a) Histogram equalization :-

Given :- $L = 4$ Total pixels = 64.

Formula: $S_k = \text{round}[(L-1) \times CDF]$

$$P(r_k) = \frac{n_k}{64}$$

Intensity	Frequency	probability	CDF	output
0	36	36/64 = 0.5625	0.5625	2
1	16	16/64 = 0.8125	0.8125	2
2	8	8/64 = 0.125	0.9375	3
3	4	4/64 = 0.0625	1	3

Final mapping :-

0→2 , 1→2 , 2→3, 3→3

b) Leaf disease image processing.

1. Image acquisition : Collect leaf images.
2. Illumination correction : Gamma correction improves brightness
3. Noise removal : Median filter removes noise.
4. Blur reduction : unsharp masking improves sharpness.
5. Background segmentation : HSV separates leaves from soil
6. Resizing : Resize images to 224 x 224
7. Normalization : Scale pixel values between 0 and 1.

8. Colour enhancement: Improves colour visibility

9. Dataset preparation: Seperate healthy and diseased images.

10. classification: CNN identifies leaf diseases.

4) Weld inspection using mean and sharpening filters.

Given Image:

$$\begin{bmatrix} 20 & 30 & 20 \\ 30 & 70 & 30 \\ 20 & 30 & 20 \end{bmatrix}$$

Step 1: Mean

$$\text{Mean} = \frac{20+30+20+30+70+30+20+30+20}{9} = \frac{270}{9}$$

$$\boxed{\text{Mean} = 30}$$

Smoothed center intensity = 30.

Step 2: Sharpening Mask

Given Mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

$$= (0 \times 20) + (-30) + 0 + (-30) + (5 \times 70) + (-30) - 30$$

$$= 350 - 120$$

$$\boxed{= 230}$$

Step 3: Difference:

Difference = original - Smooth

= 70 - 30 = 40.

comparison:-

Mean filtering	Sharpening
Reduces noise	Enhances edges
Smooth the image	Highlights fine details
Reduce local intensity variations	Increases local contrast

5) Solar panel Inspection using feature-processing technique.

Ans 1. Image acquisition: Capture solar panel image using a drone at different altitudes and viewing angles.

2. preprocessing: Convert images into grayscale and remove noise

3. Harris corner detection: Detects important corners and junctions in solar panels and identification plates.

4. Corner selection: Selects strong corners that are useful for identifying panel structures

5. Hough transform: Detects straight lines and panel boundaries.

6. Line detection: Identifies rectangular shapes and alignment of Solar Panels.

7. SIFT feature extraction: Extracts distinctive keypoints and descriptors from images.

8. Feature matching: Matches SIFT features between images captured at different scales and angles.

9. PCA (Principal Component Analysis): Reduces feature dimensions and removes redundant information

10. Final identification: Combine matched features to recognize the same Solar Panel and its identification plate.